

Investigation and Compensation of Hysteresis in Robot Joints with Cycloidal Drives

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Abstract. Improving the dynamic path accuracy has been a major research topic in industrial robotics for decades. It is known that the drivetrains installed in the robot joints limit further improvements. There is a lot more literature on the dynamic behavior of harmonic drives (HDs) than for cycloidal drives (CDs), that are usually installed in industrial robots (IRs) with heavy payload. However, a more profound knowledge of the occurring effects offers the potential for both, design- and control-based enhancements. Therefore, this paper presents an experimental study of the friction and hysteresis behavior with explicit consideration of further dependencies, such as temperature and load. Based on these investigations, a model as well as a control-based compensation approach, that does not require additional gearbox output sensors, is proposed. The investigation and validation are carried out with an experimental setup equivalent to the drivetrain of an IR with heavy payload.

Keywords: industrial robot, modeling and identification, temperature, Bouc-Wen model, sensorless compensation

1 Introduction

Since several decades, it is known that the dynamic behavior of the drivetrains installed in the robot joints are limiting the obtainable path accuracy [1,2]. These drivetrains, usually consisting of a precision gearbox and a permanent magnet synchronous machine (PMSM), exhibit a variety of nonlinear effects such as torque ripple, kinematic error, friction and hysteresis. It is also known that as precision gearboxes, CDs are installed for IRs with heavy payloads instead of HDs, which are commonly used for lightweight robots. The reason for this is the overload capability of CDs due to the operating principle with rolling contact instead of tooth meshing [3].

Previous studies [4–14] have focused almost entirely on HDs. From these investigations it is known that the friction has a dependence on additional quantities, such as temperature or load. However, to the authors best knowledge, no study has yet been published on a potential dependence of the hysteresis on additional quantities. Therefore, this paper addresses the knowledge transfer from HDs to CDs by presenting an experimental investigation of additional dependencies of

the friction and hysteresis behavior of CDs. Furthermore, a modeling of these dependencies as well as a control-based compensation approach is proposed.

2 Related Work

Since the 1990s, research efforts have been made to model the hysteresis behavior of HDs, which is caused by friction and nonlinear stiffness. Early dynamic models were proposed by Seyfferth et al. [4], Taghirad and Bélanger [5] as well as Dhaouadi et al. [6], among others. As recent work on this topic, the studies of Tjahjowidodo et al. [15] and Ruderman et al. [7,8] are noteworthy. Tjahjowidodo et al. [15] use parallel Maxwell-slip elements to describe the nonlinear dynamics, whereas Ruderman and Iwasaki [8] adopt a rate-independent Bouc-Wen hysteresis model. In addition to the modeling, Ruderman and Iwasaki propose a sensorless hysteresis compensation approach based on a generalized momentum observer [16] and a Stribeck friction model. To the authors best knowledge, only Dhaouadi et al. [6] investigated a possible multidimensionality of the hysteresis of HDs. Thereby, hysteresis curves for different frequencies were determined, and no additional dependence was found.

In contrast, there is significantly more work exploring the friction behavior. Bittencourt et al. [9] investigated the load and temperature dependence exemplarily for the second joint of an ABB IRB 6620. They detected an independence between temperature and load, and based on this, they suggested an empirical model. In contrast, in [10] no temperature, but an additional position dependence was considered using lookup tables. Carlson et al. [11] model the temperature dependence of both, an ABB IRB140 and an ABB YuMi, using a temperature-dependent Coulomb friction adaptation based on the estimated thermal energy stored in the robot joints. Simoni et al. [12] also studied the temperature dependence of the friction in the assembled state. Considering the second joint of a Comau SMART NS-16-1.65, two different modeling approaches based on a polynomial friction model were proposed. On the one hand, a model with a linear temperature dependence of the entire friction model. On the other hand, a model where each parameter exhibits an individual but linear dependence. Madsen et al. [13] consider the temperature dependence of a Universal Robot UR5e using an additive, polynomial friction term. Whereby the authors note that their approach does not extrapolate well and thus may lead to problems in practical applications. In addition, the load dependence is taken into account using an adaptation of the Coulomb friction coefficient with respect to the squared load torque. Another approach is the approximation of the temperature dependence of the friction using a neural network [14]. The validation is carried out on a testbench with a single joint of the DLRs Humanoid robot David with position, torque, and temperature sensors.

An experimental investigation of the hysteresis behavior of CDs, which is closely linked to the friction behavior, has not yet been published, unless the previous work [17]. In this work, we proposed a Bouc-Wen as well as a nonlinear

auto-regressive with exogenous inputs (NARX) model to represent the hysteresis behavior of CDs. However, a possible temperature or frequency dependence of the hysteresis behavior was not investigated. In addition, the models were not validated using a compensation scheme.

3 Experimental Setup

All subsequent investigations of the friction and hysteresis behavior of CDs are carried out on the experimental setup shown in Fig. 1, which simulates a robot joint of the heavy payload class with one degree-of-freedom. The CD under test is the precision gearbox RH380-N from Nabtesco ① with a rated torque of 3.7 kNm and a gear ratio u of 185. The lubricant temperature of the CD is measured using a PT100 sensor ⑥. The CD is driven by the PMSM MSK070D from Bosch Rexroth ②, which is equipped with a 13 Bit encoder. The joint torque τ_g is measured with the torque sensor T40B of HBM ③. Via the water-cooled high-torque motor DST2-315KO of Baumüller ⑤ a dynamic load torque can be applied. Thus, the load motor ⑤ is connected with the output-side of the CD ① using a Roba DS 1400 double-jointed coupling of Mayr ④ with a torsional stiffness of 15e6 Nm/rad.

The experimental setup is operated with a rapid prototyping platform of Speedgoat, which executes a Simulink model. The rapid prototyping platform communicates with industrial motion controllers of Bosch Rexroth and Baumüller, on which the current control of the motors run, through EtherCAT with a 2 kHz cycle time.

4 Investigation

From the related work (see Section 2) it is known that the friction has dependencies on the load and temperature in addition to the velocity. However, for the hysteresis, there is no study that examined additional dependencies. Therefore, in this section, the friction and a potential multidimensionality of hysteresis behavior of CDs is experimentally investigated.

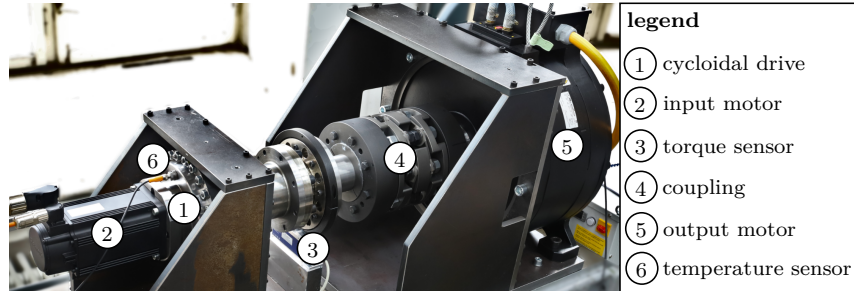


Fig. 1: Experimental setup used for the investigation (adapted from [17]).

4.1 Friction

Classical, static friction models describe a functional relationship between the friction torque τ_f and the relative velocity of the contact surfaces. Assuming only a dependence on the motor velocity $\dot{\theta}$, as is often the case in industrial robotics, it is possible to identify the friction behavior by closed-loop motion trajectories with constant velocity. We assume that the effects of temperature and load are independent, which significantly reduces the investigation burden. The validity of this assumption was already shown in [9].

To investigate an additional dependence on the load torque, this experiment was repeated several times, while a constant load torque τ_{ext} was applied using the output motor. Figure 2 shows the results for load torques of 0 to 3 kNm as well as for negative and positive loads. It is obvious that with increasing load torque, the friction torque increases, too. This can be explained by the fact that an increased load leads to an increase in the contact surface, which in turn results in a higher friction torque. However, this relationship between the load torque and the friction torque is nonlinear as well as dependent on the direction of rotation. To investigate the temperature dependence, a constant motor velocity of 150 rad/s is used to heat up the joint. Once the temperature is reached, the experiment is carried out. About 30 min were required to heat from 20 to 50 °C. The corresponding friction curves are shown in Fig. 3. The rising temperature leads to an increase in the static and Coulomb friction, whereas the viscous friction decreases. With increasing temperature, the viscosity of lubricants decreases, which explains the decrease in viscous friction. Simultaneously, the increase in temperature leads to an expansion of the material, which increases the contact surface and thus may explain the increase in static and Coulomb friction.

4.2 Hysteresis

The hysteresis behavior of robot joints is typically modeled as a nonlinear differential equation of the joint torque depending on the joint torsion as well as its derivative. Other potential dependencies such as on frequency or temperature

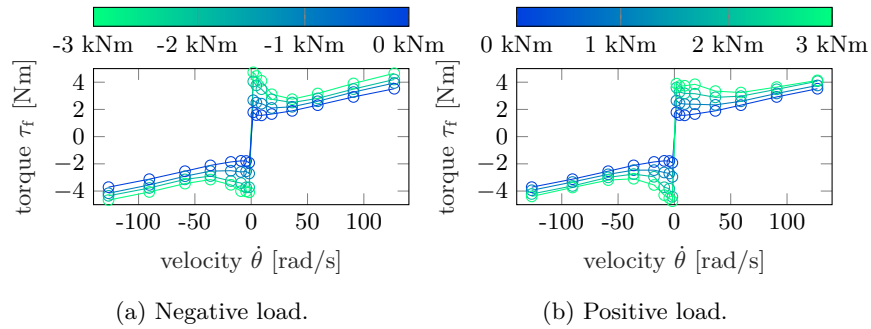


Fig. 2: Friction torque τ_f in dependence of load torque τ_{ext} .

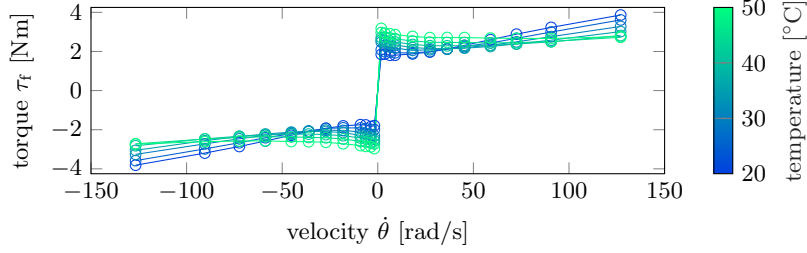


Fig. 3: Friction torque τ_f in dependence of temperature T (in steps of 5 K).

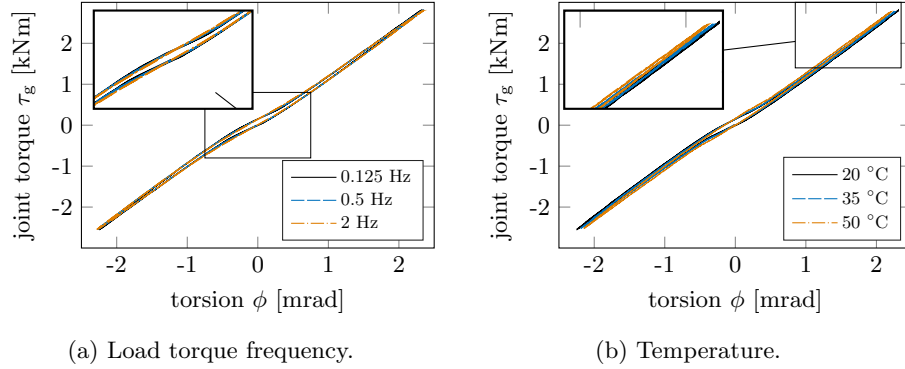


Fig. 4: Friction torque τ_f in dependence of load torque τ_{ext} frequency and temperature T .

were not examined to the authors best knowledge. The investigation of these dependencies is performed by applying a sine signal of the load torque τ_{ext} with an amplitude of 3 kNm, while varying the additional quantities. Subsequently, a static hysteresis curve is obtained in each case by plotting the joint torque τ_g against the torsion angle ϕ .

Therefore, the frequency of the sine signal of the load torque is altered between 0.125 Hz and 2 Hz. The resulting static hysteresis curves, which are nearly identical, are shown in Fig. 4a. This corresponds to a frequency independence, which is beneficial since more simple, rate-independent hysteresis models are sufficient. The procedure to heat up the robot joint corresponds to that of the investigation of the friction of Subsection 4.1. The obtained static hysteresis curve for the temperatures 20, 35 and 50 °C are shown in Fig. 4b. It is noted that a frequency of the sine signal of 0.5 Hz was chosen, however, this is irrelevant due to frequency independence. An increasing stiffness with rising temperature is noticeable, although the basic shape of the hysteresis curve does not change significantly. This stiffness increase may be explained by a temperature-dependent material expansion.

5 Compensation Method

The control-based hysteresis compensation of robot joint is an approach to meet the further increasing accuracy requirements in industrial robotics. In this case, cost-effective approaches that do not require additional gearbox output sensors are advantageous. In the following, we first propose a model based on the investigation above. Thereafter, we present a compensation approach without gearbox output sensors and validate it on the experimental setup.

5.1 Modeling

The proposed model originates from the flexible joint model according to Spong [2]. However, this model is supplemented by a temperature-dependent hysteresis spring as well as a velocity-, load-, and temperature-dependent friction. This leads to the dynamics of a single robot joint

$$\ddot{q} = M^{-1} [\tau_g + \tau_{\text{ext}}], \quad \ddot{\theta} = J_m^{-1} [\tau_m - \tau_f - u^{-1}\tau_g], \quad (1)$$

with the motor θ and joint position q , the gear ratio u , the joint-side inertia M , the motor inertia J_m , the motor τ_m , external load τ_{ext} , friction τ_f and joint torque τ_g .

For the temperature-dependent hysteresis spring we adopt a Bouc-Wen model based on our previous work [17]. This model, which is rate-independent, notes as follows:

$$\tau_g = wk\phi + (1-w)kx, \quad \dot{x} = \dot{\phi} - \beta|\dot{\phi}||x|^{n-1}x - \gamma\dot{\phi}|x|^n, \quad (2)$$

with the nonlinear, temperature-dependent stiffness

$$k(T) = k_0(T) + k_1(T)|\phi| + k_2(T)|\phi|^3, \quad (3)$$

the torsion $\phi = u^{-1}\theta - q$, the weighting factor $0 < w < 1$, the internal state x , the shape parameters γ , β , n and the temperature T . Due to the nonlinear behavior of the model, the identification is performed using the particle swarm optimization. In addition to our previous work [17], a temperature-dependent stiffness (3) is considered to account for the observed temperature behavior. Therefore, the identification procedure is repeated at 35 and 50 °C, whereas only the stiffness parameters are included as free model parameters. Subsequently, a second-order temperature-dependent polynomial is fitted separately for each stiffness parameter k_0 , k_1 , k_2 by minimizing the mean squared error (MSE) using the temperature-independent parameter estimates.

To account for the load and temperature dependence of the friction, we assume, following [9], that the temperature and load friction effects

$$\tau_f(\dot{\theta}, T, \tau_g) = \tau_{f,T}(\dot{\theta}, T) + \tau_{f,l}(\dot{\theta}, \tau_g) \quad (4)$$

are independent. For the load-dependent friction $\tau_{f,l}$ we apply a 2-D lookup table as proposed in [10]. Regarding the temperature-dependent friction $\tau_{f,T}$, we adopt a LuGre model [18]

$$\tau_{f,T} = \sigma_0 z + \sigma_1 \exp\left(-\dot{\theta}/v_d\right)^2 \dot{z} + F_v(T)\dot{\theta}, \quad \dot{z} = \dot{\theta} - \sigma_0 \left(|\dot{\theta}|/g(\dot{\theta})\right) z, \quad (5)$$

with the temperature dependent Stribeck curve

$$g(\dot{\theta}) = F_c(T) + (F_s(T) - F_c(T)) \exp\left(-|\dot{\theta}/v_s(T)|^\delta\right), \quad (6)$$

the Coulomb F_c , viscous F_v and static F_s friction coefficients, the bristle stiffness σ_0 and damping σ_1 , the shaping factor δ and the Stribeck velocity v_s . The identification is done in a two-step process. First the static friction parameters ($F_c, F_v, F_s, \delta, v_s$) are obtained using the Levenberg-Marquardt algorithm to minimize the MSE between a classical Stribeck model and the measurement (cf. Fig. 3) at each temperature. Secondly, for each of parameter separately, a second order temperature-dependent polynomial is fitted in the same way as for the hysteresis behavior. Subsequently, the dynamic parameters σ_0, σ_1, v_d are identified employing a particle swarm optimization.

5.2 Compensation Scheme

The proposed compensation scheme, which is shown in Fig. 5, is adapted from [8, 19]. The compensation is based on an inversion of the hysteresis, requiring the joint torque, which is not measured. Instead of a sensor, a so-called generalized momentum observer, which is known from collision detection of robots [16], is utilized. The starting point for the derivation is the generalized momentum $p = J_m \cdot \dot{\theta}$. The observer yields by taking the time derivative

$$\dot{p} = J_m \cdot \ddot{\theta} = \tau_m - \tau_f - u^{-1} \tau_g \quad (7)$$

and the residual

$$r = K_o \left[\int \dot{p} dt - p \right] = K_o \left[\int (\tau_m - \tau_f - r) dt - p \right] \quad (8)$$

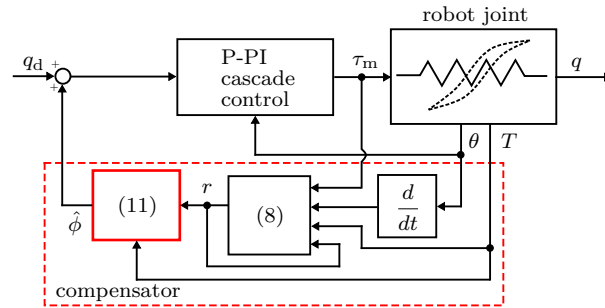


Fig. 5: Control scheme of hysteresis compensation approach.

of the generalized momentum. This residual equals the estimate of the joint torque, which becomes obvious by taking its time derivative

$$\dot{r} = K_o [\tau_m - \tau_f - r - \dot{p}] = K_o [-r + u^{-1}\tau_g] \quad (9)$$

and transforming it into the Laplace domain

$$\lim_{K_o \rightarrow \infty} r/u^{-1}\tau_g = \lim_{K_o \rightarrow \infty} K_o/s + K_o = 1. \quad (10)$$

Subsequently, the hysteresis behavior (2) is inverted to obtain the estimated joint torsion

$$\hat{\phi} = 1/wk \cdot [ur - (1 - w)kx]. \quad (11)$$

Finally, the estimated torsion is added to the desired joint position q_d .

It is known from [19], that residual oscillations may occur using this compensation scheme. To avoid this effect, a dead zone of the position error is included, which matches the noise of the estimated joint torsion $\hat{\phi}$ at standstill.

Moreover, the compensation scheme needs an estimate of the friction. In addition to [8, 19] we apply a LuGre observer instead of a static Stribeck model. The observer

$$\hat{\tau}_{f,T} = \sigma_0 \hat{z} + \sigma_1 \exp\left(-\dot{\theta}/v_d\right)^2 \hat{z} + F_v \dot{\theta} \quad (12a)$$

$$\dot{\hat{z}} = \dot{\theta} - \sigma_0 \left(|\dot{\theta}|/g(\dot{\theta})\right) \hat{z} + k_f \left(\tau_m - J_m \ddot{\theta}_d - r - (\hat{\tau}_{f,T} + \hat{\tau}_{f,l})\right), \quad (12b)$$

with the observer gain k_f , is adapted from [20] and extended by the previously modeled temperature and load dependence. Due to an insufficient sensor resolution, we apply the desired $\ddot{\theta}_d$ instead of the measured motor acceleration $\ddot{\theta}$.

5.3 Experimental Validation

The experimental validation is performed on the test bench of Fig. 1 by applying a point-to-point trajectory of the desired joint position q_d with a trapezoidal velocity profile. Simultaneously, a sinusoidal load torque τ_{ext} with an amplitude of 3 kNm and a frequency of 1/16 Hz, imitating a gravity induced force, is set. The experiment is conducted at a gearbox lubricant temperature of $T = 20$ and 35°C . Figure 6 shows the desired joint position q_d , load torque τ_{ext} and tracking errors $e_q = q_d - q$ of the experiment. The presented tracking errors correspond to the scenarios without compensation e_q , with compensation e_c according to [8] and the proposed compensation scheme e_e . With the compensation according to [8], the $\mathcal{L}_1$ norm of the tracking error at $T = 20^\circ\text{C}$ is reduced by 81 % from 13.8 mrad to 2.64 mrad. However, oscillations are evident at standstill. Using the proposed compensation, the $\mathcal{L}_1$ norm is reduced by another 56 % to 1.17 mrad as well as the oscillations are avoided. At the temperature of $T = 35^\circ\text{C}$, the tracking error is reduced to a larger extent regarding the compensation according to [8] due to the modeled temperature dependence.

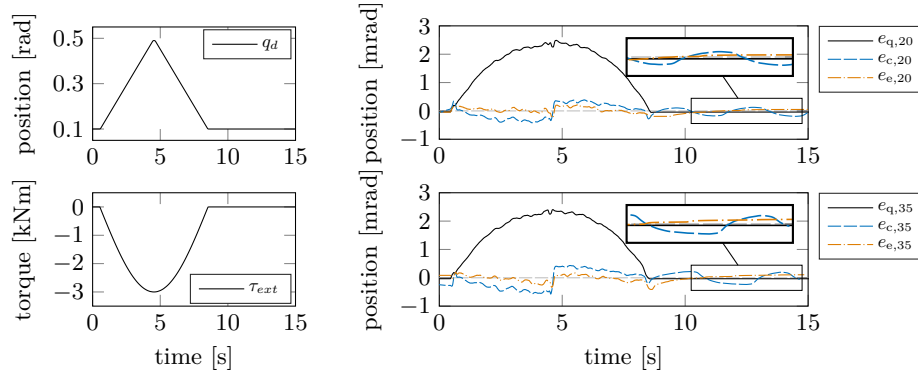


Fig. 6: Tracking experiment without (black), with compensation according to [8] (blue) and the proposed compensation (yellow) at $T = 20$ and 35 °C.

6 Conclusions

In this paper, an experimental investigation of the friction and hysteresis behavior of cycloidal drives was presented. The investigation revealed a significant load and temperature dependence of the friction. However, the hysteresis is rate-independent, and there is a low, temperature-dependent increase in stiffness. Therefore, the results indicate a great similarity between HDs and CDs regarding the friction and hysteresis behavior. Moreover, a compensation approach with an extended friction model was proposed, which improves the trajectory tracking performance compared to the state of the art.

In the future the temperature sensor may be replaced by an observer. In addition the approach should be validated on a six degree-of-freedom manipulator in a practical application such as milling.

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References

1. Sweet, L., Good, M.: Re-definition of the robot motion control problem: Effects of plant dynamics, drive system constraints, and user requirements. In: The 23rd IEEE Conference on Decision and Control. pp. 724–732. IEEE (1984)
2. Spong, M.W.: Modeling and control of elastic joint robots. Journal of Dynamic Systems, Measurement, and Control 109(4), 310–318 (1987)
3. Sensinger, J.W., Lipsey, J.H.: Cycloid vs. harmonic drives for use in high ratio, single stage robotic transmissions. In: IEEE International Conference on Robotics and Automation. pp. 4130–4135. IEEE, Piscataway, NJ (2012)
4. Seyffarth, W., Maghzal, A.J., Angeles, J.: Nonlinear modeling and parameter identification of harmonic drive robotic transmissions. In: IEEE International Conference on Robotics and Automation. pp. 3027–3032. IEEE, Piscataway, NJ (1995)

5. Taghirad, H.D., Be'langer, P.R.: Modeling and parameter identification of harmonic drive systems. *Journal of Dynamic Systems, Measurement, and Control* 120(4), 439–444 (1998)
6. Dhaouadi, R., Ghorbel, F.H., Gandhi, P.S.: A new dynamic model of hysteresis in harmonic drives. *IEEE Transactions on Industrial Electronics* 50(6), 1165–1171 (2003)
7. Ruderman, M., Hoffmann, F., Bertram, T.: Modeling and identification of elastic robot joints with hysteresis and backlash. *IEEE Transactions on Industrial Electronics* 56(10), 3840–3847 (2009)
8. Ruderman, M., Iwasaki, M.: Sensorless torsion control of elastic-joint robots with hysteresis and friction. *IEEE Transactions on Industrial Electronics* 63(3), 1889–1899 (2016)
9. Bittencourt, A.C., Wernholt, E., Sander-Tavallaey, S., Brogårdh, T.: An extended friction model to capture load and temperature effects in robot joints. In: *The IEEE/RSJ International Conference on Intelligent Robots and Systems*. pp. 6161–6167. IEEE, Piscataway, N.J (2010)
10. Wahrburg, A., Klose, S., Clever, D., Groth, T., Moberg, S., Styrd, J., Ding, H.: Modeling speed-, load-, and position-dependent friction effects in strain wave gears. In: Lynch, K. (ed.) *2018 IEEE International Conference on Robotics and Automation*. pp. 2095–2102. IEEE, Piscataway, NJ (2018)
11. Carlson, F.B., Robertsson, A., Johansson, R.: Modeling and identification of position and temperature dependent friction phenomena without temperature sensing. In: *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. pp. 3045–3051. IEEE (2015)
12. Simoni, L., Beschi, M., Legnani, G., Visioli, A.: Modelling the temperature in joint friction of industrial manipulators. *Robotica* 37(5), 906–927 (2019)
13. Madsen, E., Rosenlund, O.S., Brandt, D., Zhang, X.: Comprehensive modeling and identification of nonlinear joint dynamics for collaborative industrial robot manipulators. *Control Engineering Practice* 101, 104462 (2020)
14. Wolf, S., Iskandar, M.: Extending a dynamic friction model with nonlinear viscous and thermal dependency for a motor and harmonic drive gear. In: Lynch, K. (ed.) *2018 IEEE International Conference on Robotics and Automation (ICRA)*. pp. 783–790. IEEE, Piscataway, NJ (2018)
15. Tjahjowidodo, T., Al-Bender, F., van Brussel, H.: Theoretical modelling and experimental identification of nonlinear torsional behaviour in harmonic drives. *Mechatronics* 23(5), 497–504 (2013)
16. de Luca, A., Mattone, R.: Actuator failure detection and isolation using generalized momenta. In: *2003 IEEE International Conference on Robotics and Automation (Cat. No.03CH37422)*. pp. 634–639. IEEE (2003)
17. Mesmer, P., Nagel, P., Lechler, A., Verl, A.: Modeling and identification of hysteresis in robot joints with cycloidal drives. In: *2022 IEEE 17th International Conference on Advanced Motion Control (AMC)*. pp. 358–363. IEEE (2022)
18. Olsson, H., Åström, K.J., Canudas de Wit, C., Gäfvert, M., Lischinsky, P.: Friction models and friction compensation. *European Journal of Control* 4(3), 176–195 (1998)
19. Ruderman, M., Iwasaki, M.: Impact of hysteresis lost motion on the sensorless torsion control of elastic robotic joints. In: *2015 IEEE International Conference on Mechatronics (ICM)*. pp. 654–659. IEEE (2015)
20. Freidovich, L., Robertsson, A., Shiriaev, A., Johansson, R.: Lugre-model-based friction compensation. *IEEE Transactions on Control Systems Technology* 18(1), 194–200 (2010)